

화학수학

아홉 번째 수업

적분변환

Functional Series (함수급수)

$$s(x) = a_0 g_0(x) + a_1 g_1(x) + a_2 g_2(x) + \cdots$$

- Coefficients (계수): $a_0, a_1, a_2, \cdots$
 - Basis functions (바탕함수): $g_0(x), g_1(x), g_2(x), \cdots$
1. Power series (멱급수): powers as basis
 2. Fourier series (푸리에 급수): sines & cosines as basis

Fourier Series

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \sum_{n=0}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=0}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

- **Periodicity:** $f(x) = f(x + 2L) = f(x + 4L) = \dots$
- **Completeness:** the series can exactly represent any function

Orthogonality

The basis functions are orthogonal to each other:

$$\int_{-L}^L \cos\left(\frac{m\pi x}{L}\right) \cos\left(\frac{n\pi x}{L}\right) dx = \begin{cases} 2L, & \text{if } m = n = 0 \\ L\delta_{mn}, & \text{otherwise} \end{cases}$$

$$\int_{-L}^L \sin\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx = \begin{cases} 0, & \text{if } m = n = 0 \\ L\delta_{mn}, & \text{otherwise} \end{cases}$$

$$\int_{-L}^L \cos\left(\frac{m\pi x}{L}\right) \sin\left(\frac{n\pi x}{L}\right) dx = 0$$

Fourier Series Coefficients

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$a_0 = \frac{1}{2L} \int_{-L}^L f(x) \, dx$$

$$a_{n \neq 0} = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) \, dx$$

$$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) \, dx$$

Fourier Series Coefficients

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

	Odd $f(x)$	Even $f(x)$
$a_0 = \frac{1}{2L} \int_{-L}^L f(x) dx$	0	
$a_{n \neq 0} = \frac{1}{L} \int_{-L}^L f(x) \cos\left(\frac{n\pi x}{L}\right) dx$	0	
$b_n = \frac{1}{L} \int_{-L}^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$		0

sine series cosine series

Complex Exponential Basis Functions

$$a_n \cos\left(\frac{n\pi x}{L}\right) + b_n \sin\left(\frac{n\pi x}{L}\right) = \frac{a_n - ib_n}{2} e^{in\pi x/L} + \frac{a_n + ib_n}{2} e^{-in\pi x/L}$$

The following two formulas are equivalent:

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{L}\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right)$$

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/L}$$

Orthogonality and Coefficients

- Orthogonality

$$\int_{-L}^L (e^{im\pi x/L})^* e^{in\pi x/L} dx = 2L\delta_{mn}$$

- Coefficients

$$c_m = \frac{1}{2L} \int_{-L}^L f(x) e^{-im\pi x/L} dx$$

General Functional Series

- Complete set of orthogonal basis functions:

$$\psi_1, \psi_2, \dots, \psi_n, \dots$$

- Expansion of an arbitrary function f :

$$f = \sum_n c_n \psi_n$$

- Hilbert space: $\psi_1, \psi_2, \dots, \psi_n, \dots$ span the Hilbert space

Scalar Products in the Hilbert Space

- The scalar product of $f(x)$ and $g(x)$ is

$$\int f^*(x)g(x) dx$$

- The scalar product of $f(x, y, z)$ and $g(x, y, z)$ is

$$\iiint f^*(x, y, z)g(x, y, z) dx dy dz$$

Orthogonality and Normalization

- The basis functions are **orthogonal**:

$$\int \psi_n^* \psi_m \, dx = c \delta_{nm}$$

- They can be **normalized**:

$$\int \psi_n^* \psi_m \, dx = \delta_{nm}$$

- We assume the normalized basis functions

Finding the Coefficients

$$f(x) = \sum_n c_n \psi_n$$

Multiplying both sides by ψ_m^* and integrating,

$$\int \psi_m^* f dx = \sum_n c_n \int \psi_m^* \psi_n dx$$

$$\int \psi_m^* f dx = \sum_n c_n \delta_{mn} = c_m$$

Today's Class

- Fourier transforms
 - Fourier series and Fourier transforms
 - Some variations
- Laplace transforms
 - Laplace transforms
 - Shifting, derivative, and integral theorems



Integral Transforms



Fourier transform

Laplace transform

Mellin transform

Hankel transform

⋮

Fourier Transforms

- Fourier series

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/L}$$

$$k = \frac{n\pi}{L}, L \rightarrow \infty$$

- Fourier transform (= Fourier integral)

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(k) e^{ikx} dk$$

How to determine?

Another Fourier Transforms

- Coefficient determination

$$c_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-in\pi x/L} dx$$

- Fourier transform

$$F(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-ikx} dx$$

$$k = \frac{n\pi}{L}, L \rightarrow \infty$$


Transform and Inverse Transform

- Fourier transform

$$F(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-ikx} dx$$

- Inverse Fourier transform

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} F(k) e^{ikx} dk$$

Example 11.4

Find the Fourier transform of the Gaussian function

$$f(x) = e^{-ax^2}.$$

Using the definition of the Fourier transform,

$$F(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-ikx} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ax^2} e^{-ikx} dx$$

Since $-ax^2 - ikx = -a(x^2 + ikx/a) = -a(x + ik/2a)^2 - k^2/4a$,

$$\begin{aligned} F(k) &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-a(x+ik/2a)^2} e^{-k^2/4a} dx = \frac{1}{\sqrt{2\pi}} e^{-k^2/4a} \int_{-\infty}^{\infty} e^{-a(x+ik/2a)^2} dx \\ &= \frac{1}{\sqrt{2\pi}} e^{-k^2/4a} \sqrt{\frac{\pi}{a}} = \frac{1}{\sqrt{2a}} e^{-k^2/4a} \end{aligned}$$

Fourier Cosine and Sine Transforms

- Fourier cosine transform: for an **even** function $f(x)$,

$$F(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) \cos(kx) dx$$

- Fourier sine transform: for an **odd** function $f(x)$,

$$F(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) \sin(kx) dx$$

One-Sided Transforms

- One-sided Fourier transform ($x > 0$)

$$F(k) = \left(\frac{2}{\pi}\right)^{1/2} \int_0^{\infty} f(x) e^{-ikx} dx$$

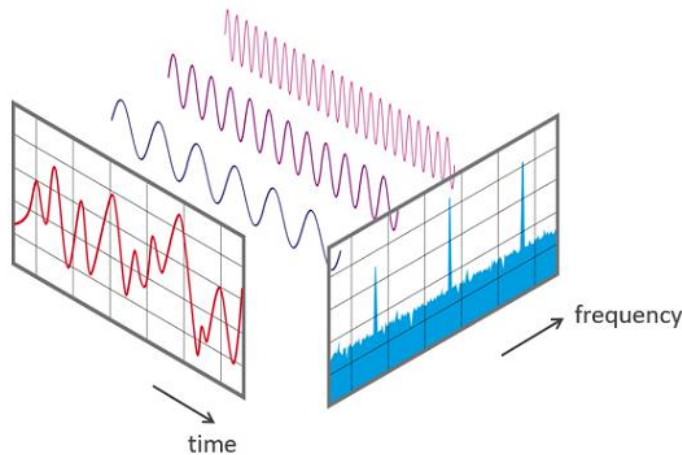
- One-sided inverse Fourier transform ($k > 0$)

$$f(x) = \left(\frac{2}{\pi}\right)^{1/2} \int_0^{\infty} F(k) e^{ikx} dk$$

Application: Time $\leftrightarrow$ Frequency Transform

$$F(\omega) = \left(\frac{2}{\pi}\right)^{1/2} \int_0^{\infty} f(t) e^{-i\omega t} dt$$

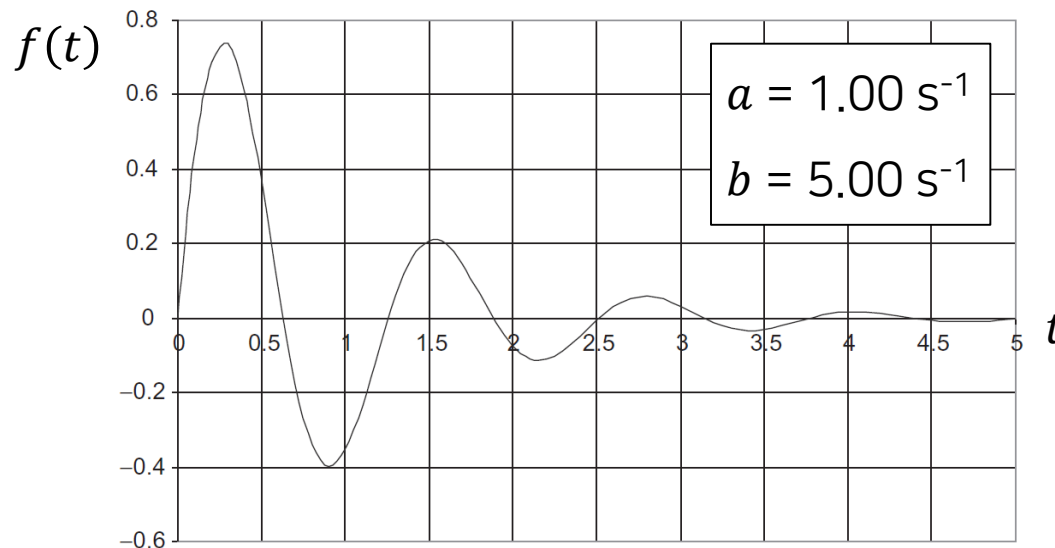
$$f(t) = \left(\frac{2}{\pi}\right)^{1/2} \int_0^{\infty} F(\omega) e^{i\omega t} d\omega$$



(Image: <https://www.nti-audio.com/en/support/know-how/fast-fourier-transform-fft>)

Example 11.5

Find the one-sided sine Fourier transform of the “ringing exponential” function $f(t) = e^{-at} \sin(bt)$.



Example 11.5

Find the one-sided sine Fourier transform of the “ringing exponential” function $f(t) = e^{-at} \sin(bt)$.

Using the definition of the Fourier transform,

$$F(\omega) = \left(\frac{2}{\pi}\right)^{1/2} \int_0^{\infty} f(t) \sin(\omega t) dt = \left(\frac{2}{\pi}\right)^{1/2} \int_0^{\infty} e^{-at} \sin(bt) \sin(\omega t) dt$$

Using the table of integrals,

$$F(\omega) = \left(\frac{2}{\pi}\right)^{1/2} \frac{2ab\omega}{[a^2 + (b - \omega)^2][a^2 + (b + \omega)^2]}$$

Some Common Fourier Transforms

$f(x)$	$F(k)$	$f(x)$	$F(k)$
1	$\sqrt{2\pi}\delta(k)$	$e^{-a x }$	$\left(\frac{2a^2}{\pi}\right)^{1/2} \frac{1}{k^2 + a^2}$
e^{iax}	$\sqrt{2\pi}\delta(k - a)$	$xe^{-a x }$	$-\left(\frac{2a^2}{\pi}\right)^{1/2} \frac{2ik}{(k^2 + a^2)^2}$
e^{-iax}	$\sqrt{2\pi}\delta(k + a)$	$e^{-a^2x^2}$	$\frac{1}{\sqrt{2}a} e^{-k^2/4a^2}$
$1/x$	$-i\left(\frac{\pi}{2}\right)^{1/2} \text{sgn}(k)$	$\frac{1}{x^2 + a^2}$	$\left(\frac{\pi}{2a^2}\right)^{1/2} e^{-a k }$

- Dirac delta function $\delta(x) = \begin{cases} \infty & \text{if } x = 0 \\ 0 & \text{if } x \neq 0 \end{cases}$
- Sign function $\text{sgn}(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases}$

Laplace Transforms

- (One-sided) Fourier transform

$$F(\omega) = \left(\frac{2}{\pi}\right)^{1/2} \int_0^{\infty} f(t) e^{-i\omega t} dt$$

- Laplace transform

$$F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

Example 11.7

Find the Laplace transform of the function $f(t) = t^2$.

From the definition of the Laplace transform,

$$F(s) = \int_0^{\infty} f(t)e^{-st} dt = \int_0^{\infty} t^2 e^{-st} dt$$

By setting $u = st$ and using the table of integrals,

$$F(s) = \frac{1}{s^3} \int_0^{\infty} u^2 e^{-u} du = \frac{2!}{s^3} = \frac{2}{s^3}$$

Inverse Laplace Transforms

- Laplace transform

$$F(s) = \int_0^{\infty} f(t)e^{-st} dt = \mathcal{L}\{f(t)\}$$

- Inverse Laplace transform

$$f(t) = \frac{1}{2\pi i} \int_{c-i\infty}^{c+i\infty} F(s)e^{st} ds = \mathcal{L}^{-1}\{F(s)\}$$

requires complex analysis;
We will not use this formula

Some Common Laplace Transforms

$F(s)$	$f(t)$
$1/s$	1
$1/s^2$	t
$n!/s^{n+1}$	t^n
$\frac{1}{s-a}$	$e^{at} (s > a)$
$\frac{s}{s^2 + a^2}$	$\cos(at)$
$\frac{a}{s^2 + a^2}$	$\sin(at)$
$\frac{s}{s^2 - a^2}$	$\cosh(at)$
$\frac{a}{s^2 - a^2}$	$\sinh(at)$

Shifting Theorem

$$\mathcal{L}\{e^{at}f(t)\} = F(s - a)$$

Example 11.8

Use the shifting theorem to obtain the Laplace transform of the function

$$f(t) = e^{at} \cos(kt).$$

From Table 11.1,

$$\mathcal{L}\{\cos(kt)\} = \frac{s}{s^2 + k^2}$$

Hence, from the shifting theorem, $\mathcal{L}\{e^{at}f(t)\} = F(s - a)$,

$$\mathcal{L}\{e^{at} \cos(kt)\} = \frac{s - a}{(s - a)^2 + k^2}$$

Derivative Theorem and Integral Theorem

- Derivative theorem

$$\mathcal{L}\left\{\frac{df(t)}{dt}\right\} = s\mathcal{L}\{f(t)\} - f(0)$$

- Integral theorem

$$\mathcal{L}\left\{\int_0^t f(u) du\right\} = \frac{1}{s}\mathcal{L}\{f(t)\}$$

Example 11.9

Find the inverse Laplace transform of

$$F(s) = \frac{1}{s(s-a)}.$$

From Table 11.1,

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$$

The integral theorem states that

$$\mathcal{L}\left\{\int_0^t f(u) \, du\right\} = \frac{1}{s} \mathcal{L}\{f(t)\}$$

Hence,

$$F(s) = \frac{1}{s(s-a)} = \frac{1}{s} \mathcal{L}\{e^{at}\} = \mathcal{L}\left\{\int_0^t e^{au} \, du\right\}$$

Example 11.9

Find the inverse Laplace transform of

$$F(s) = \frac{1}{s(s-a)}.$$

$$F(s) = \mathcal{L} \left\{ \int_0^t e^{au} \, du \right\}$$

The inverse gives

$$\mathcal{L}^{-1}\{F(s)\} = \int_0^t e^{au} \, du = \frac{1}{a} e^{au} \Big|_0^t = \frac{1}{a} (e^{at} - 1)$$